Computer Communication and Networks



Network Layer

Session 13

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Network Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Define Network Address Translation (NAT)
- Identify Types of NAT
- Explain Transition Strategies



Network Layer: NAT

Address Depletion

- With the usage of the internet rising exponentially, IPv4 addresses were depleting.
  - Therefore, a short term solution to the problem of the depletion of IP addresses was devised:
    1. CIDR
    2. NAT

  - What is long time solution to this issue?
    - ✔ IPv6



Network Layer: NAT

- The organization responsible for registering IP address ranges to organizations and Internet Service Providers (ISPs) is **Internet Assigned Numbers Authority** (IANA).
  - To allow organizations to freely assign private IP addresses, the Network Information Center (InterNIC) has reserved certain address blocks for private use.

    The following IP blocks are reserved for private IP addresses.

    10.0.0.0 – 10.255.255.255

    172.16.0.0 – 172.31.255.255

    192.168.0.0 – 192.168.255.255

These IPs can be used within a network, campus, or company and are private to it.
These addresses cannot be routed on the Internet, so packets containing these private addresses are dropped by the Routers.



Network Layer: NAT

Rules for assigning Network ID

- The network ID will be assigned based on the below-given rules:
  1. The network ID cannot start with 127 because 127 belongs to the class A address and is reserved for internal **loopback functions**.
  2. All bits of network ID set to **1** are reserved for use as an IP broadcast address and cannot be used.
  3. All bits of network ID is set to **0**. They are used to denote a particular host on the local network and should not be routed.



Network Layer: NAT

Loopback IP addresses

- It is also called a localhost address
- The IP address range 127.0.0.0 – 127.255.255.255 is reserved for loopback
- This loopback IP address is managed entirely by and within the **operating system**.
- Loopback addresses, enable the **Server** and **Client** processes on a single system to communicate with each other.



Network Layer: NAT

NAT

- Two types of addresses
  1. Public IP addresses
  2. Private IP addresses.



Network Layer: NAT

Public IP Addresses

- A **public IP address** is the address that is assigned to a computing device to allow **direct access** over the **Internet**.
- A **web server**, **email server**, and any server device directly accessible from the Internet.
- A **public IP address** is **globally unique**, and can only be assigned to a unique device.



Network Layer: NAT

Private IP Addresses

- Private IP network is an IP network that is not directly connected to the **Internet**.
  - When a computer is assigned a **private IP address**, the local devices see this computer via its private IP address.
  - However, the devices residing outside of your local network cannot directly communicate via the private IP address, but uses your router's public IP address to communicate.
  - To allow direct access to a local device that is assigned a private IP address, a Network Address Translator (NAT) should be used.



Network Layer: NAT

Network Address Translation

- **NAT** is an **Internet standard** that enables a Local Area Network (LAN) to:
  - ✔ Use one set of IP addresses for internal traffic and
  - ✔ A second set of IP addresses for external traffic.

- A NAT box located where the LAN meets the Internet makes all necessary IP address translations

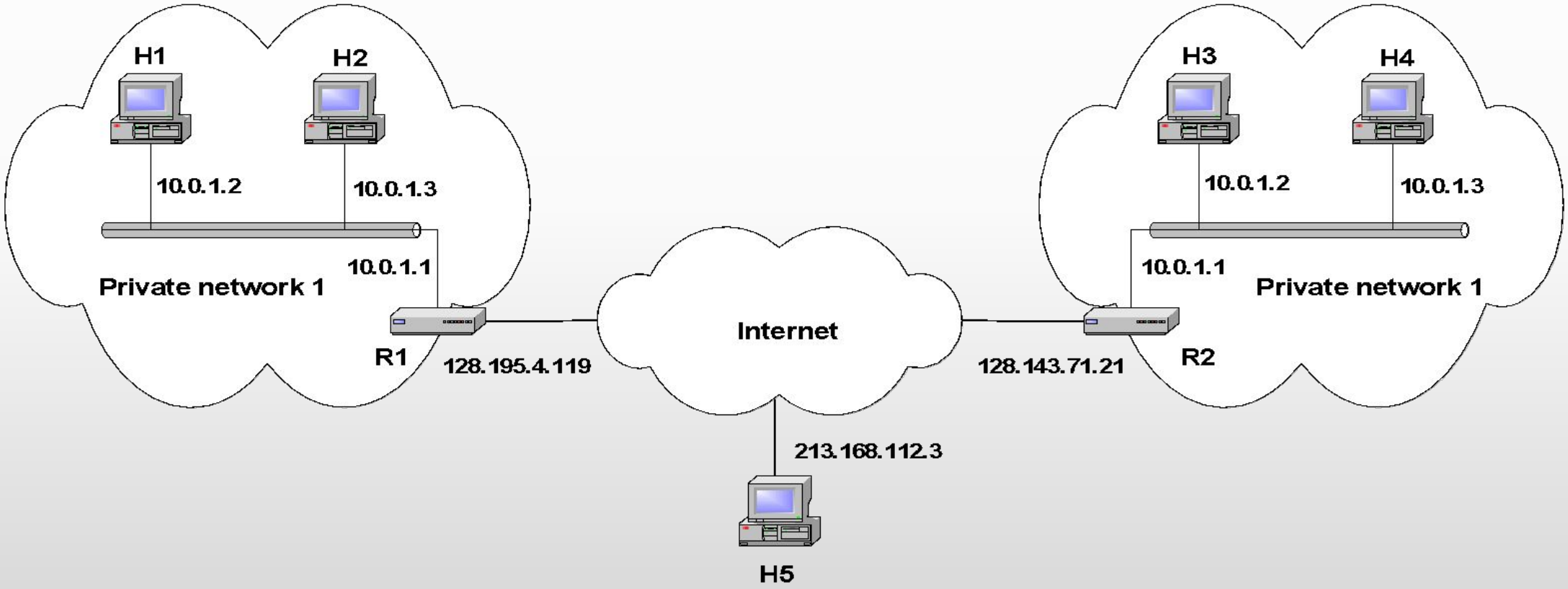


Network Layer: NAT

Network Address Translation

- The Purpose of NAT
  - NAT serves three main purposes:
    1. Provides a type of **firewall** by hiding internal IP addresses
    2. Enables a company to use more internal IP addresses.
       - Since they're used internally only, there's no possibility of conflict with IP addresses used by other companies and organizations.
    3. Allows a company to combine multiple ISDN connections into a single Internet connection.

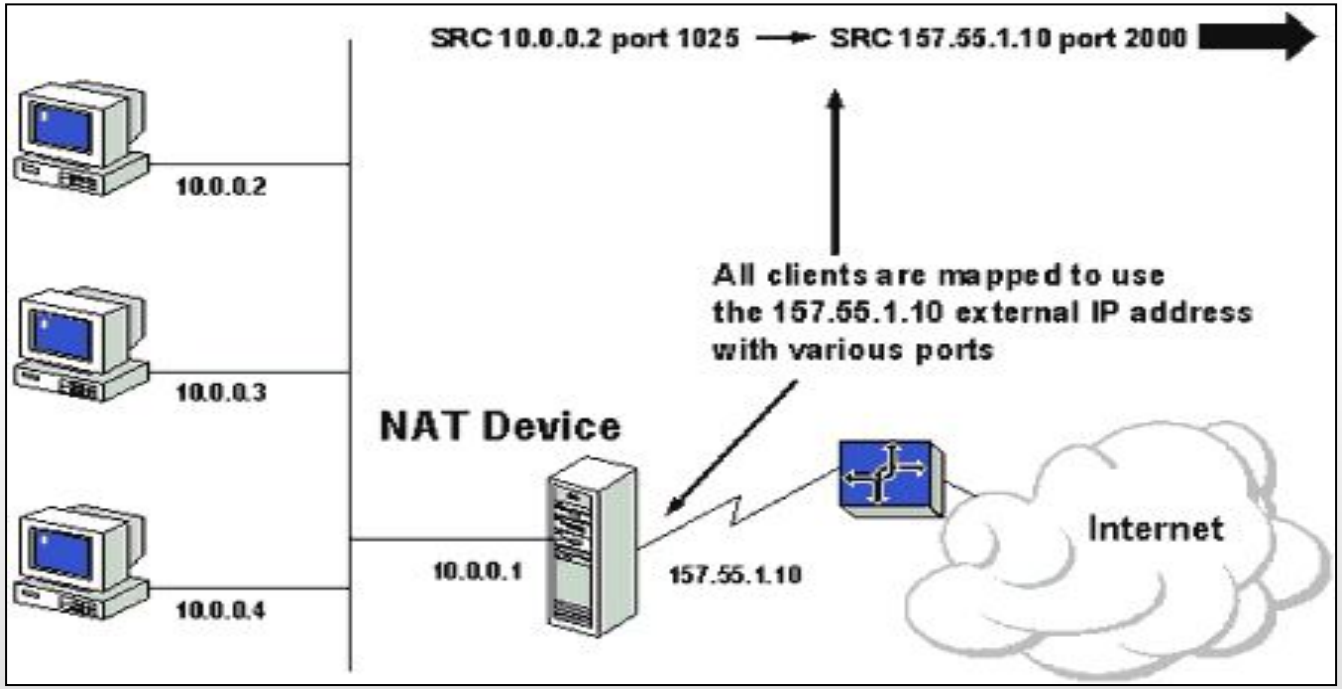
Network Address Translation: NAT





Network Layer: NAT

Network Address Translation: NAT





Network Layer: Network Address Translation

NAT: What is NAT?

- Network Address Translation translates the **private addresses** into **public addresses** before packets are routed to public network.
  - ✔ It allows a network device such as Router to translate addresses between the private and public network
- Situation in which NAT is required:
  1. When we need to connect to Internet and our hosts doesn't have globally unique IP addresses.
  2. When we want to hide internal IP addresses from outside for security purpose.
  3. A company is going to merge in another company which uses same address space.



Network Layer: Network Address Translation

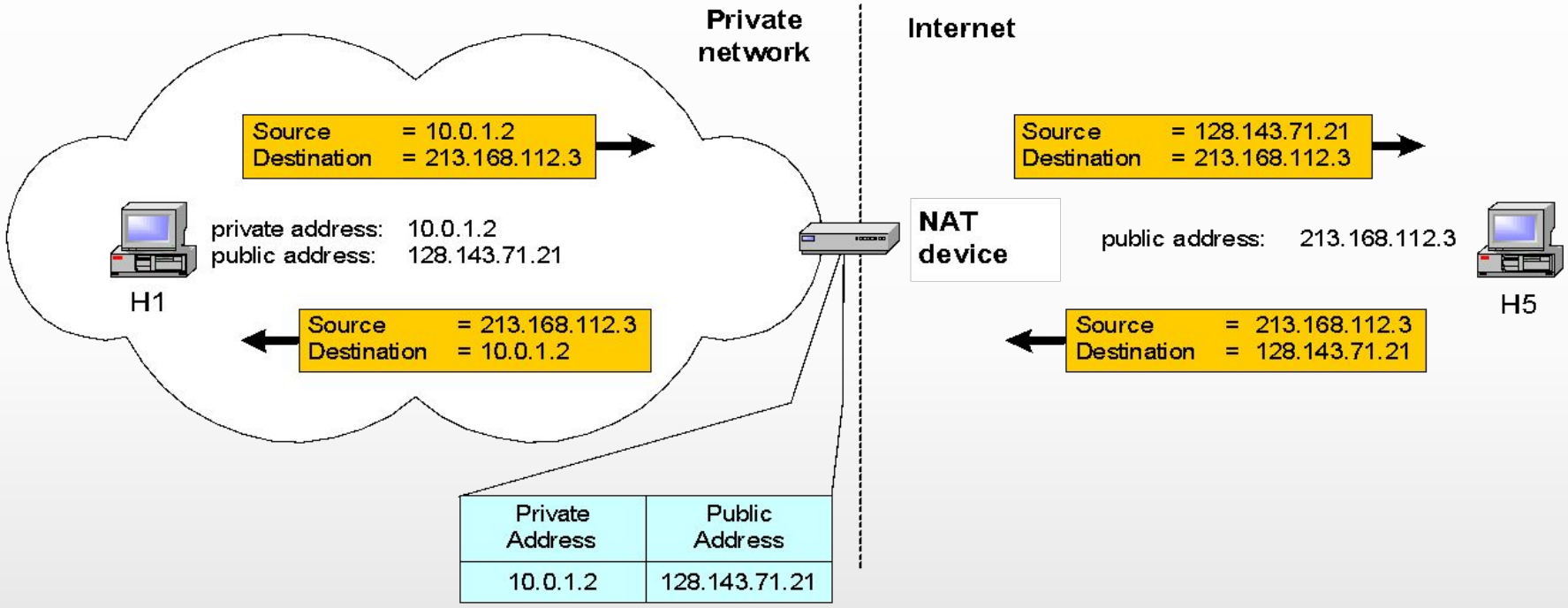
NAT: Advantages of NAT

1. It conserves legally registered IP addresses.
2. It prevents address overlapping.
3. Provides security by hiding internal (private) IP addresses.
4. Eliminates address renumbering as a network evolves.



Network Layer: Network Address Translation

NAT: Basic Operation of the NAT



- NAT device has address translation table
  - ✔ One to one address translation



NAT: Types of NAT

- Three different types of NAT are:
  1. Static NAT
  2. Dynamic NAT
  3. Port Address Translation (Overloading)

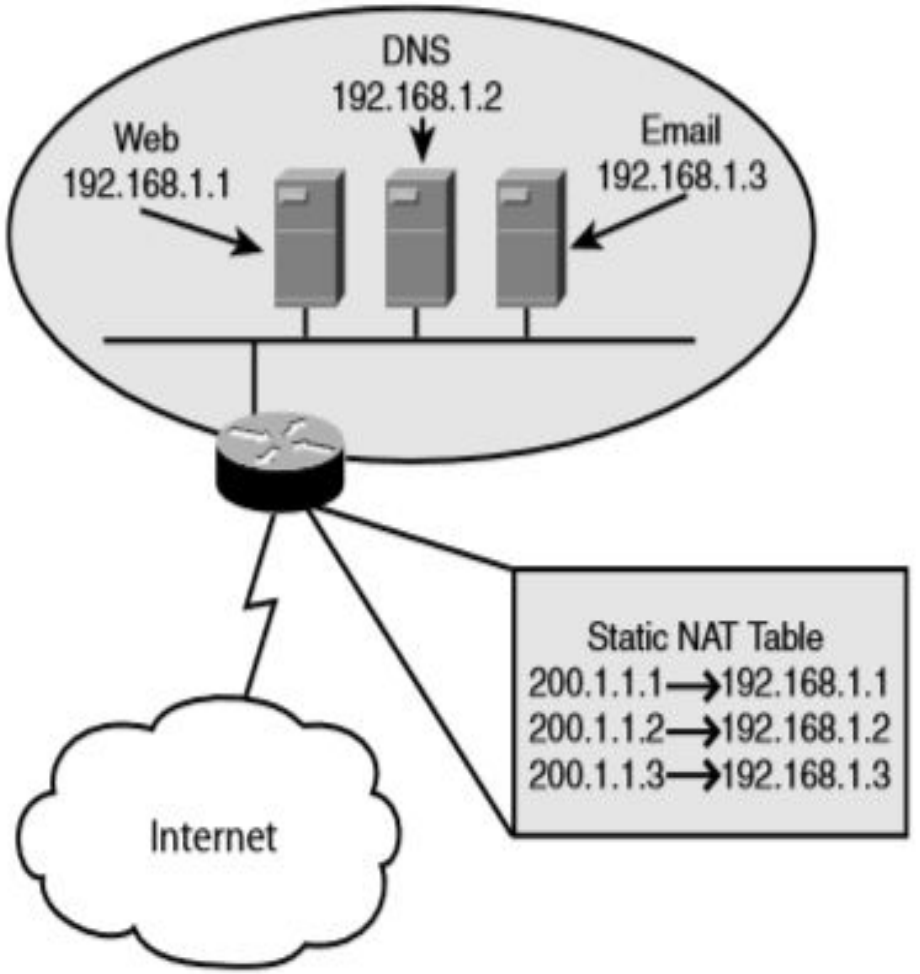
- What is Static NAT?
  - ✔ Static NAT allows for one to one mapping
    - That is, it translates one Private IP address to one Public IP address.



Network Layer: Network Address Translation

NAT: Static NAT

- This Static NAT mapping goes both ways.
  - ✔ When someone from the Internet accesses 200.1.1.1, it is translated to the internal address 192.168.1.1.
  - ✔ Likewise, when the server 192.168.1.1 accesses the Internet, it is seen as 200.1.1.1.





NAT: Dynamic NAT

- This is where dynamic NAT can help.
  - Dynamic NAT enables you to define a **pool of addresses** to be translated along with a pool of addresses to which they are to be translated.

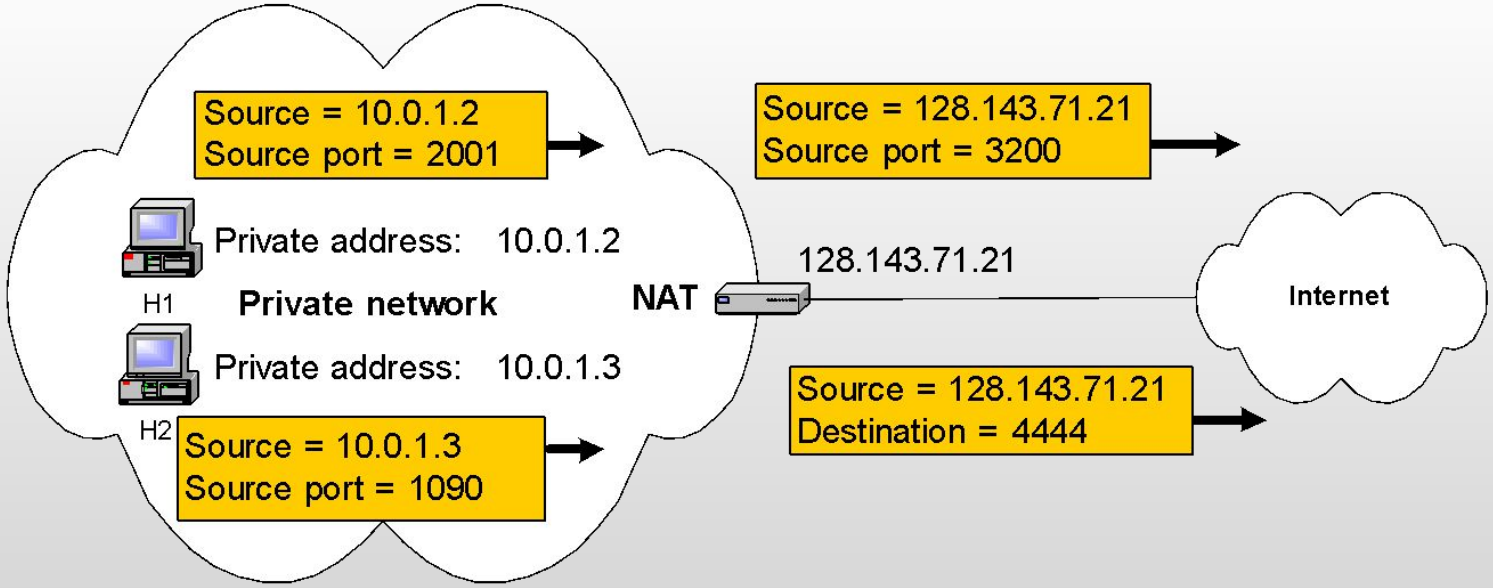
- What is Dynamic NAT?
  - **Static** NAT is superb if you have a **few hosts** that need to be translated;
  - However, if many hosts need to be translated, creating static entries for each one can be quite tedious.



Network Layer: Network Address Translation

NAT: Port Address Translation PAT

- PAT (Port Address Translation)
  - Maps a Public IP and Port Number to a service in Private IP
- ✔ Maps Multiple Private IP to a Pool of Public IPs
- ✔ PAT allows thousands of users to connect to Internet using one public address only.





Port Address Translation

- Most home networks use PAT.
  - In such a scenario, the Internet Service Provider (ISP) assigns a single IP address to the home network's router.
    - When Computer X logs on the Internet, the router assigns the client a port number, which is appended to the **internal IP address**.
      - ✔ This, in fact, gives Computer X a unique address.
    - If Computer Z logs on the Internet at the same time, the router assigns it the same local IP address with a different **port number**.
    - Although both computers are sharing the same public IP address and accessing the Internet at the same time, the router knows exactly which computer to send specific packets to because each computer has a unique internal address.



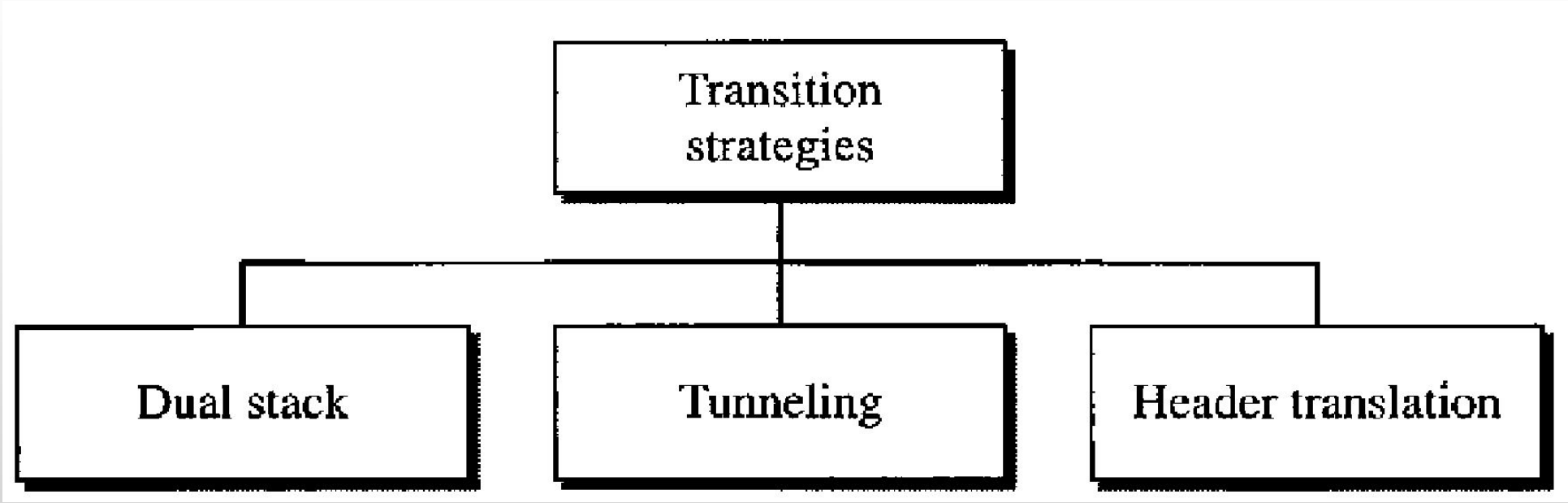
Transition Strategies



Network Layer: Transition

Transition from IPv4 to IPv6:

- Transition from IPv4 to IPv6 cannot happen suddenly. Why?
  - ✔ Huge number of systems on the Internet.
- Smooth transition – To prevent any problems between IPv4 and IPv6 systems.
- **Three transition strategies**:

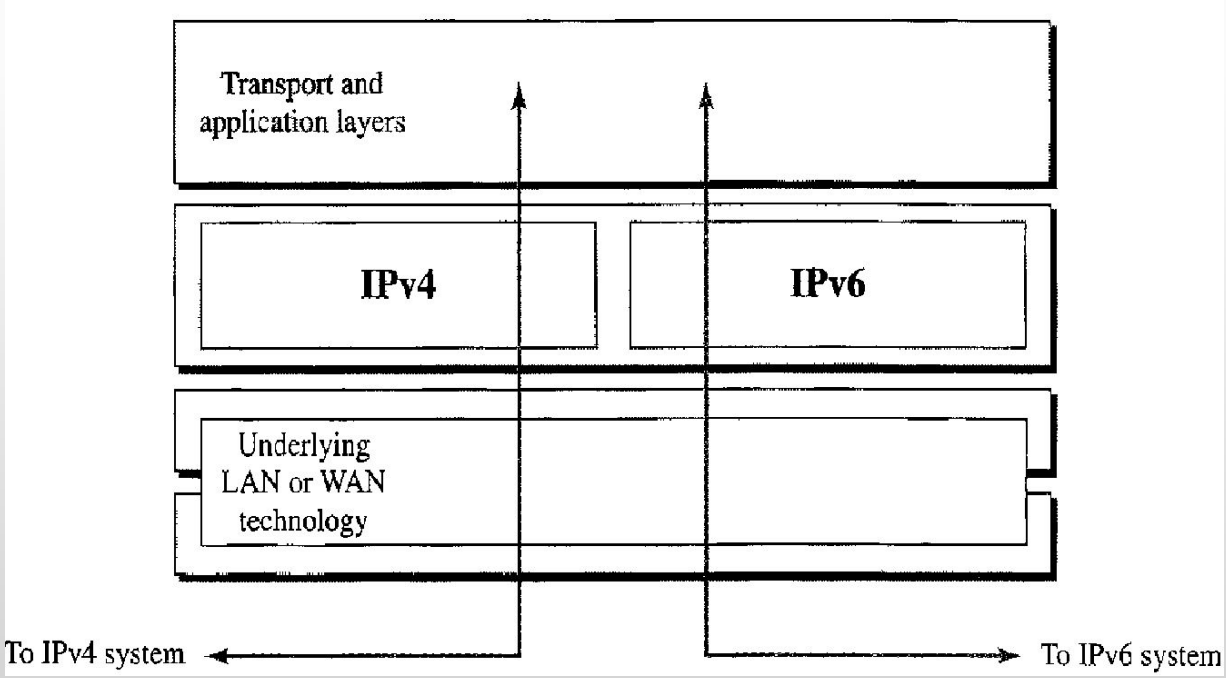




Network Layer: Transition Strategies

Dual Stack:

- Both IPv4 and IPv6 protocols
- **A station** must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.
- Source station queries the DNS to get the version. Then send the IPv4/IPv6 packet after getting reply from the DNS.

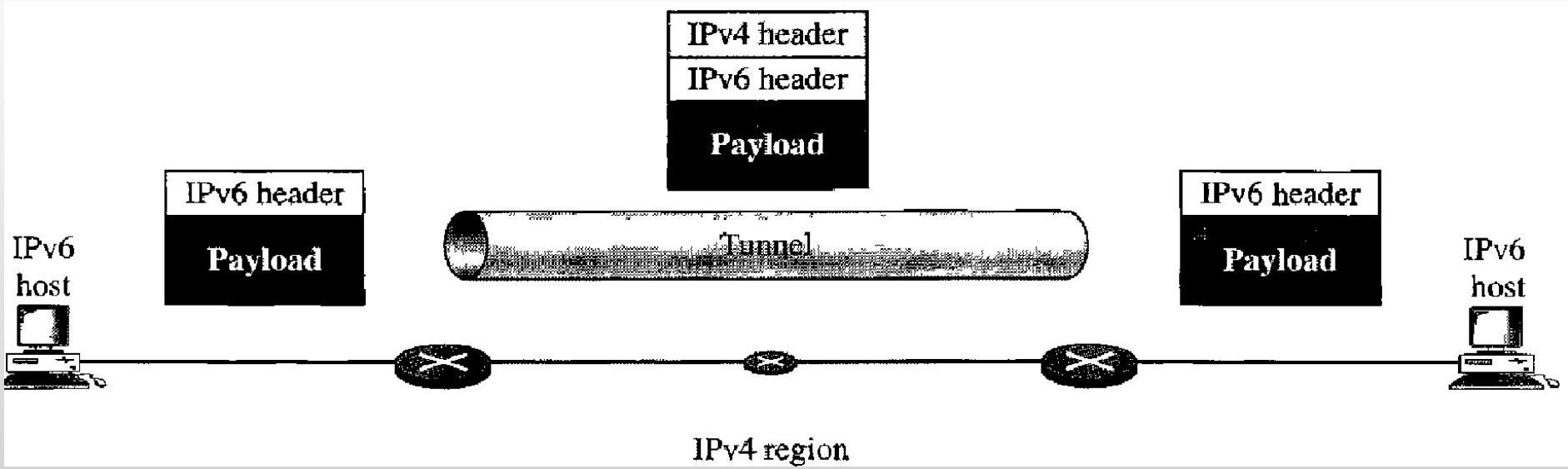




Network Layer: Transition Strategies

Tunneling:

- When two computers using IPv6 wants to communicate each other and packet must pass through region that uses IPv4.
- Therefore, packet must have IPv4 address. So IPv6 packet is encapsulated in an IPv4 packet when enters into the region and leaves its capsule when exits the region.

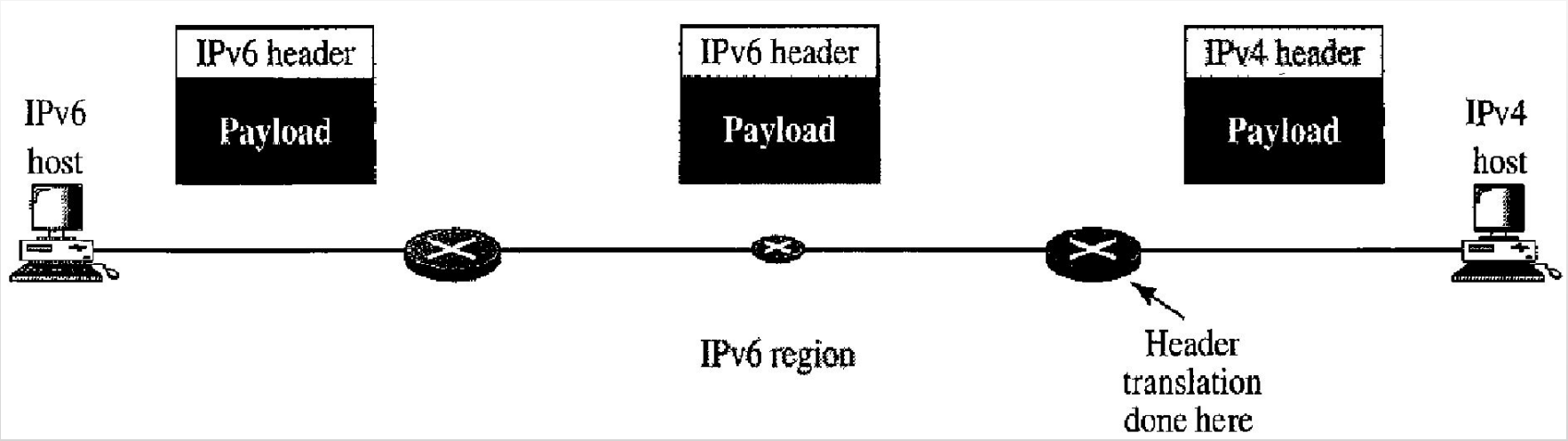




Network Layer: Transition Strategies

Header Translation:

- This strategy is also necessary when the majority of the Internet has moved to IPv6.
  - ✔ Problem? Some systems still use IPv4
- The sender wants to use IPv6, but the receiver doesn't understand IPv6 and tunneling does not work in this situation. So packet must be in IPv4 format and header format must totally changed to IPv4.





References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan